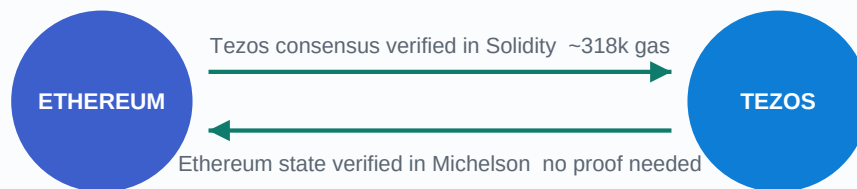


Ethereum and Tezos, verifying each other. No bridge.

Art preserved as provable bytes — not a token, not a pointer, not a promise.

Almost everything built between chains moves tokens through a bridge you have to trust. Stratus does the opposite: it makes the artwork itself provable, and lets each chain check the other's proof on its own — no oracle, no attester set, nobody in the middle.



The same artwork exists on both. Independently checkable on each.

If one chain fails, the other still proves it. Bitcoin works too, as a third copy.

WHY THIS IS DIFFERENT

- **Not token-based.** The proof is about bytes, not ownership. Attach it to an NFT, or keep it standalone as a backup that outlives the token, the marketplace, and the company that minted it.
- **Not a bridge.** Nothing is wrapped or moved. Each chain confirms the content itself, so there is no bridge to hack and no failure that propagates.
- **Not maximalist.** Ethereum and Tezos are both first-class. The goal is the same work living in both places at once, not an argument about which chain wins.

Both directions, measured

Against live mainnet data, not simulations.

Tezos -> Ethereum

~318k gas · no prover

Solidity verifies Tezos BLS consensus signatures natively, using the BLS12-381 precompiles added in Pectra.

Ethereum -> Tezos

no prover

Michelson walks Ethereum's Merkle-Patricia state trie directly, using native KECCAK. Verified against a real mainnet storage proof.

Tezos quorum -> Ethereum

20.6M cycles

77 signatures across three schemes, proven in a zkVM because 35M gas of Ed25519 will not fit in a block.

Bitcoin -> both

third copy

A real ordinal inscription verified on Ethereum and Tezos, reaching the identical anchor root by independent means.

THE ECONOMICS OF ONE HARD CASE

Three of those four need no proof at all — each chain simply does the arithmetic. The exception is Tezos consensus on Ethereum: about half of Tezos validators sign with Ed25519 and half with P-256, and the EVM has a precompile for neither. Checking it directly would cost roughly 35 million gas, more than an Ethereum block holds.

Proved instead, it verifies for about 300 thousand. That gap is the entire economic case for a verification layer, and it is why zkVerify's aggregation is the cheapest route we found: a Merkle check rather than a pairing check, amortised across everyone else's proofs.

Where zkVerify fits

STATUS: INTEGRATION COMPLETE, PROOF BUILT, NOT YET SUBMITTED

Our verification route calls zkVerify's live aggregation contract on Sepolia, with the leaf digest transcribed from zkV-attestation-contracts and confirmed ABI-compatible against the deployed proxy. A proof is built and formatted for submission; the Volta account is still short of the fee, so no receipt exists yet. Until one does, we describe this as built and wired rather than proven in production.

WHAT ZKVERIFY VERIFIES FOR US

Anchor proofs on the Ethereum side. When an artwork is anchored, a proof establishes that a foreign chain really holds those bytes, and something on Ethereum has to check it. That check is the route zkVerify serves: our contract reconstructs the statement digest, then asks zkVerify's aggregation contract whether that statement sits in a published aggregation.

On the Tezos side the arithmetic is done directly in Michelson, which has native BLS12-381 pairing and KECCAK. Two destinations, two verification routes, one architecture — the proof-checking layer is deliberately pluggable so each chain uses whatever is cheapest there.

WHY AGGREGATION IS THE RIGHT SHAPE FOR THIS

Verifying a proof outright costs a pairing check. Verifying its membership in an aggregation costs a Merkle check, with the expensive part amortised across every other proof in the batch. For a workload like ours that difference compounds, and the measurements say why: in our Bitcoin proof the chain-of-headers walk is 5,859 cycles per header and everything else is a flat 140k, so the expensive part is shared by every artwork proven against the same segment. Batching them into one proof is designed, not yet built.

Latency is irrelevant to us. These are permanent records, not trades. Nobody is waiting on a confirmation, so batching costs us nothing and saves us the per-proof price.

WHY THE STORY TRAVELS

Verification layers are always explained through rollups and throughput. This is a different shape: cultural preservation, where the value is that a work can still be proven real long after whoever minted it is gone. Same technology, an audience that has never heard of it.

Every figure here is measured against mainnet or explicitly marked as not yet done.